Date : / /

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AIUB, Physics 1 final term

#final TERM

Topic Name : Day : lesson 8.1
Time : Date : / /

Table of Contents		
SL.NO	**Topic**	**Pages**

Topic Name :

Day : lesson 8.1

Time :   Date : / /

Ch: 22 Electric Fields

Topic Name : Day : Time : Date : / /

Electric Fields:—

⇒ The electric field at a certain point is equal to the electric force per unit charge experienced by a charge at the point.

$$\boxed{\vec{E} = \frac{\vec{F}}{q_0}}$$

SI Unit is $= NC^{-1}$

* if q_0 charge gives = F force

so, 1 " " " $= \frac{F}{q_0}$ "

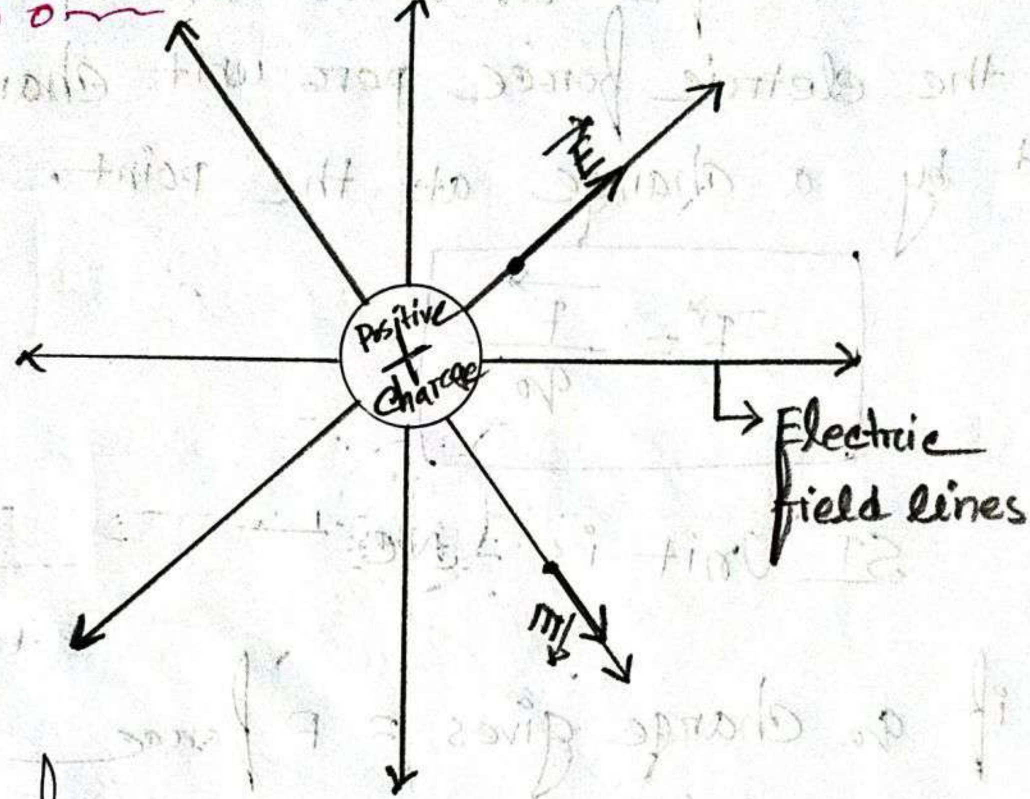
Q:- What is 1 unit positive charge.?

⇒ 1 unit positive charge is the amount of electric charge carried by a single proton or equivalent to 1.602×10^{-19} coulombs.

Topic Name :   Day :
                                                Time :   Date : / /

Electric field lines around charged particles:—

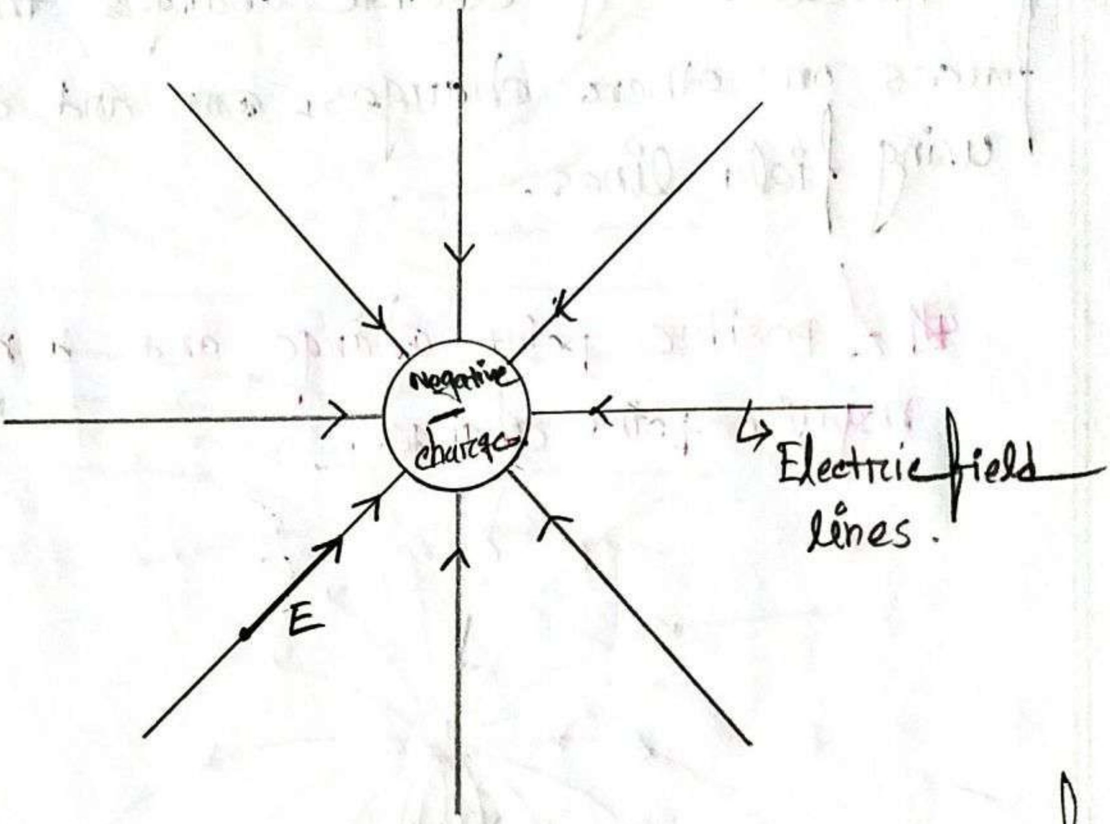
for positive:—



⇒ the field produced by a positive point charge points away from the charge.

Topic Name : Day : Time : Date : / /

for negative charges:-



⇒ The field produced by a negative point charge points toward the charge.

Notes:-

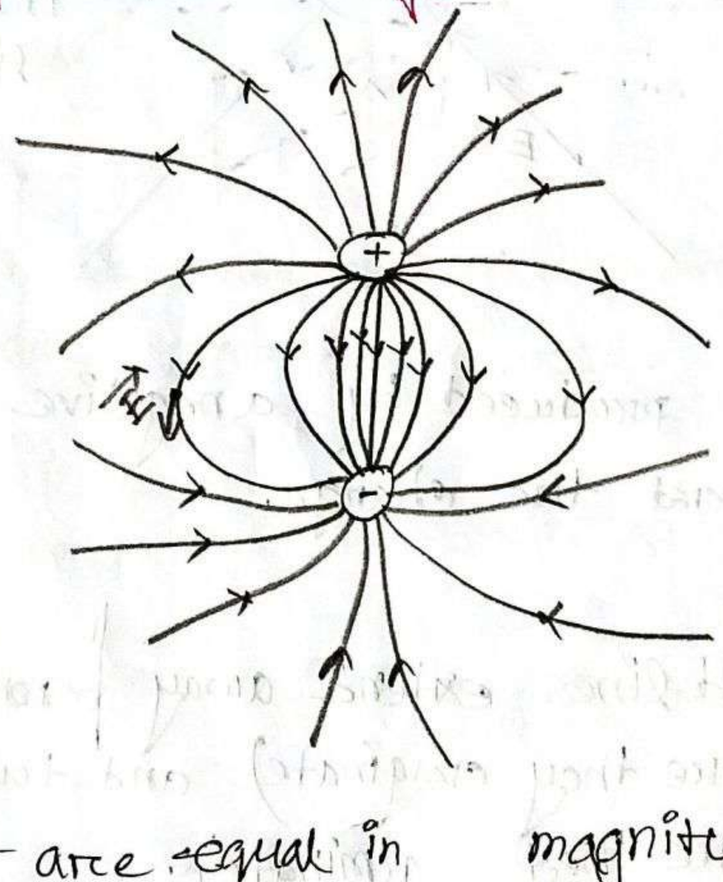
→ Electric field lines extend away from positive charge (where they originate) and toward negative charge (where they terminate).

Topic Name : Day : Time : Date : / /

Properties of Electricfield:-

☒ Electric fields are a vector quantities Produced by electric charges that exert forces on other charges, and can be visualized using field lines.

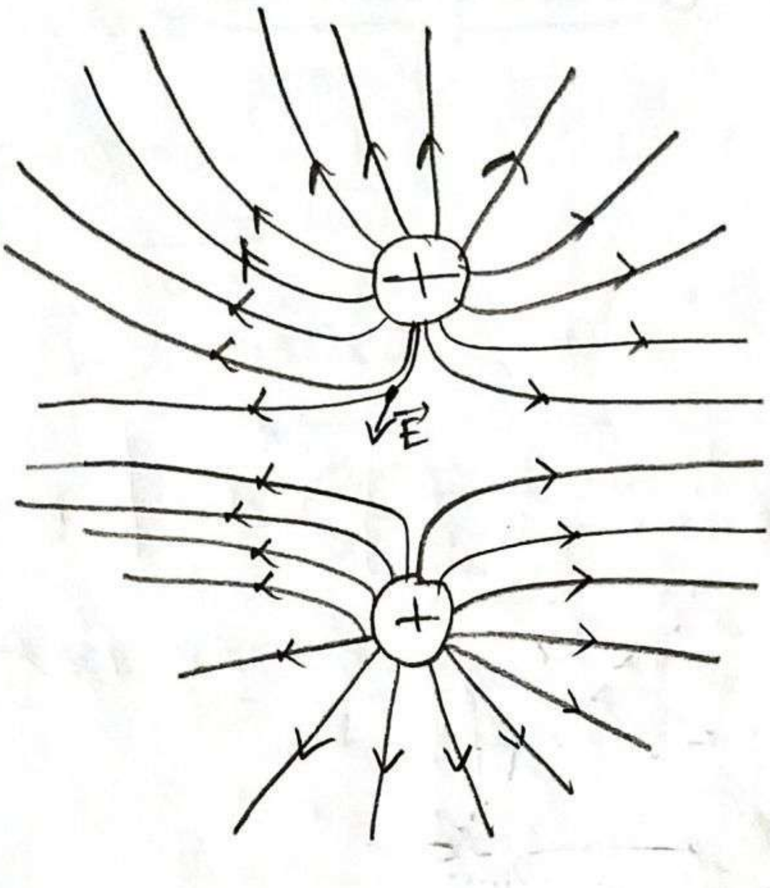
☒ A positive point charge and a nearby negative point charge.



→ That are equal in magnitude.

Topic Name : ____________ Day : ______ Time : Date : / /

Two equal positive point charges.



Notes—

→ Electric field lines help us visualize the direction and magnitude of electric field.

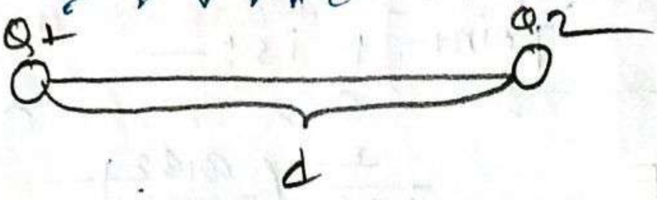
The electric field vector at any point is tangent to the field line through that point. The density of field lines in that region is proportional to the magnitude of the electric field there. Thus, closer field lines represent a strong field.

Topic Name :

Day :

Time : Date : / /

State of colombs law:



$$F \propto Q_1 \times Q_2$$

$$F \propto \frac{1}{d^2}$$

So,

$$F \propto \frac{Q_1 \times Q_2}{d^2}$$

$$F = K \frac{Q_1 Q_2}{d^2}$$

where, $K = \frac{1}{4\pi\epsilon_0 \epsilon_r}$

Now

$$F = \frac{Q_1 Q_2}{4\pi\epsilon_0\epsilon_r d^2}$$

Electric field due to a point charge:

$$\boxed{F = \frac{1}{4\pi\epsilon_0} \times \frac{q_1 q_2}{r^2}}$$

$$\boxed{\epsilon_0 = 8.85 \times 10^{-12}\ C^2 N^{-1} m^{-2}}$$

Topic Name: Day: Time: Date: /

The magnitude E of the electric field at point p is:—

$$E = \frac{F}{q_2} = \frac{\frac{1}{4\pi\epsilon_0}\left(\frac{q_1 q_2}{r^2}\right)}{q_2}$$

$$= \frac{1}{4\pi\epsilon_0} \times \frac{q}{r^2}$$

So, $\boxed{\vec{E} = \frac{1}{4\pi\epsilon_0} \times \frac{q}{r^2} \hat{r}}$ → Vector form

$\boxed{E = \frac{1}{4\pi\epsilon_0} \times \frac{q}{r^2}}$ SI unit NC^{-1}

Q: Problem 5:—

A charged particle produces an electric field with a magnitude of $2NC^{-1}$ at a point that is 50 cm away from the particle. What is the magnitude of the particle's charge?

⇒ is given,

$E = 2NC^{-1}$
$r = 50 \times 10^{-2}$

Topic Name : Day : Time : Date : / /

$$\frac{1}{4\pi\epsilon_0} = \frac{1}{4\pi \times 8.85\times10^{-12}} = 9\times10^{9}\ Nm^2/C^2$$

We know,

$$E = \frac{1}{4\pi\epsilon_0}\times\frac{q}{r^2}$$

$$\Rightarrow q = \frac{E}{9\times10^9}\times r^2$$

$$= \frac{2}{9\times10^9}\times(50\times10^{-2})^2$$

$$= 5.556\times10^{-11}\ C$$

Q: What is the magnitude of a point charge that would create an electric field of 1.00 N/C at points 1.00 m away?

$\Rightarrow$ is given,

$E = 1\ NC^{-1}$

$r = 1\ m$

We know, $E = K\times\frac{q}{r^2}$

$$\Rightarrow q = \frac{Er^2}{K}$$

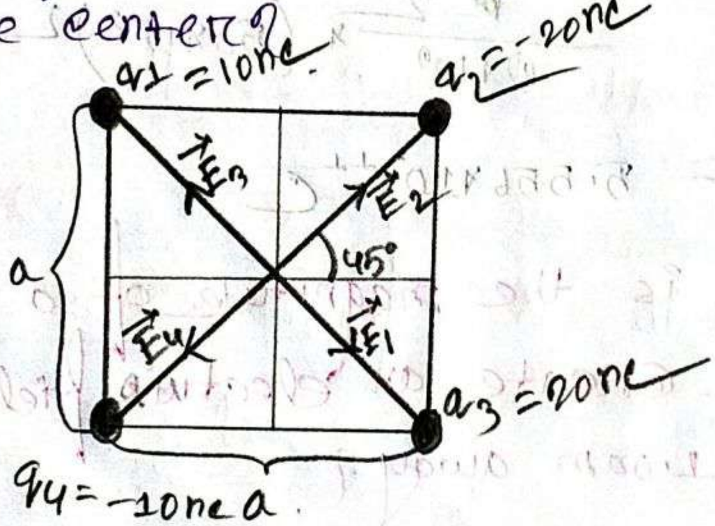
$$= \frac{1\times1^2}{9\times10^9}$$

$$= 1.11\times10^{-10}\ C$$

Topic Name : Day : Time : Date : / /

Problem: 7

In the adjacent figure, the four particles form a square of edge length $a = 5.00cm$ and have a charge $q_1 = +10nC$, $q_2 = -20nC$, $q_3 = +20nC$, and $q_4 = -10nC$. In unit vector notation, what net electric field do the particles produce at the square center?



The net electric field at the center of the square along the x-axis is :-

$$E_x = E_2 \cos 45^\circ + E_1 \cos 45^\circ - (E_3 \cos 45^\circ + E_4 \cos 45^\circ) \quad \text{———} \quad (I)$$

here,

$$E_1 = E_4 = \frac{1}{4\pi\epsilon_0}\left(\frac{10\times10^{-9}}{r^2}\right) \quad \text{———} \quad (III)$$

$$E_2 = E_3 = \frac{1}{4\pi\epsilon_0}\left(\frac{20\times10^{-9}}{r^2}\right) \quad \text{———} \quad (IV)$$

Topic Name : Day : Time ; Date : / /

Subsitude the Value of E_3 and E_4 in equation (1)

$E_x = E_2 \cos 45^\circ + E_1 \cos 45^\circ - (E_2 \cos 45^\circ + E_1 \cos 45^\circ)$

$= E_2 \cos 45 - E_2 \cos 45 + E_1 \cos 45 - E_1 \cos 45$

$= 0$

The net electric field at the center of the square along Y axis is:—

$E_y = E_2 \cos 45^\circ + E_3 \cos 45 - (E_1 \cos 45 + E_4 \cos 45^\circ)$ —— (II)

Subsitude the Value of E_3 and E_4 in equation (II) $\Rightarrow$

$E_y = E_2 \cos 45^\circ + E_2 \cos 45 - (E_1 \cos 45^\circ + E_1 \cos 45^\circ)$

$= 2E_2 \cos 45 - 2E_1 \cos 45^\circ$

$= 2 \times \frac{1}{4\pi\epsilon_0} \times \frac{q_2}{r^2} \times \cos 45^\circ - 2 \times \frac{1}{4\pi\epsilon_0} \times \frac{q_1}{r^2} \cos 45^\circ$

$= 2 \times \frac{1}{4\pi\epsilon_0} \times \frac{1}{r^2} \times \cos 45^\circ (q_2 - q_1)$

$= 2 \times 9 \times 10^9 \times \frac{1}{(0.0354)^2} \times \cos 45 (20 \times 10^{-9} - 10 \times 10^{-9})$

$= 101.566 \times 10^3$ N/C

$qp^2 = a^2 + a^2$

$qp = \sqrt{a^2 + a^2}$

$qp = \sqrt{2a^2}$

$op = \frac{\sqrt{2 \times (5 \times 10^{-2})^2}}{2}$

$= 0.0354$

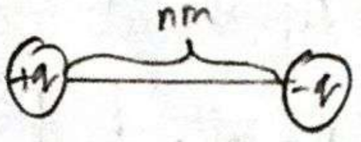
total $\vec{E} = E_x \hat{i} + E_y \hat{j} = 0 + 101.566 \times 10^3 \hat{j}$

$= 101.566 \times 10^3 \hat{j}$ N/C Ans

Topic Name : Day : lessons 2 Time : Date : / /

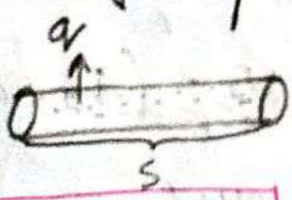
Dipoles

⇒ A dipole is a pair of equal and opposite electric charge or magnetic poles separated by a small distance.



Linear charge density

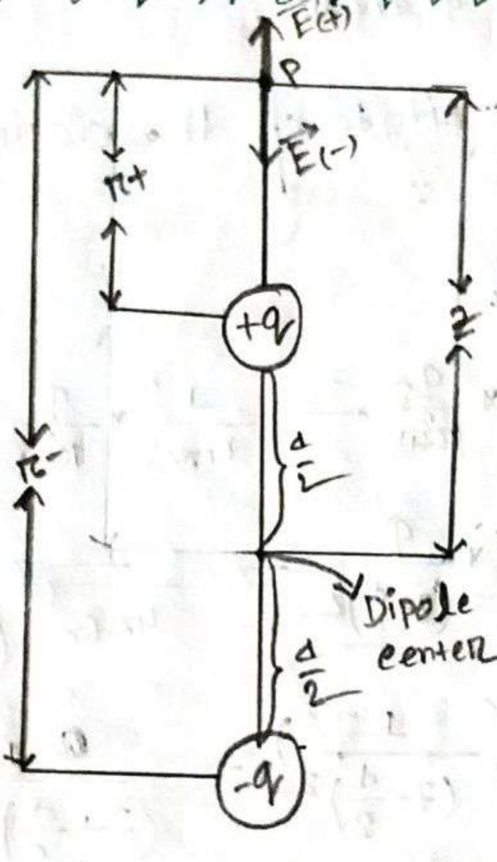
⇒ Linear charge density is the amount of electric charge per unit length of a one dimensional object.



$$\boxed{\lambda = \frac{q}{s}}$$

Electric fields:—

Electric field for a elictric Dipole:—



we all know the noramal equation is:—

$$E = \frac{1}{4\pi\epsilon_0} \times \frac{q}{r^2}$$

⇒ [কিন্তু ডাইপোল / দ্বিমেরুর ক্ষেত্রে কিন্তু কেন্দ্র থাকে এই Normal Equation এর মাধ্যমে কাজ হবে এখানে Double charge থাকবে ও হবে, এখন আমরা দেখব এর জন্য equation বের করবো।] (FP)

P.T.O

Topic Name : Day : Time : Date : / /

$r_+ = z - \frac{d}{2}$

$r_- = z + \frac{d}{2}$

& $E_+ \rightarrow$ for $+q$ charge.

$E_- \rightarrow$ for $-q$ charge.

So, the net m gnitude of the electric field at point P is:-

$$E = E_{(+)} - E_{(-)}$$

$$= \frac{1}{4\pi\epsilon_0} \times \frac{q}{r^2_{(+)}} - \frac{1}{4\pi\epsilon_0} \times \frac{q}{r^2_{(-)}}$$

$$= \frac{1}{4\pi\epsilon_0} \times \frac{q}{(z-\frac{d}{2})^2} - \frac{1}{4\pi\epsilon_0} \frac{q}{(z+\frac{d}{2})^2}$$

$$= \frac{q}{4\pi\epsilon_0} \left\{ \left(\frac{1}{(z-\frac{d}{2})^2} \right) - \frac{1}{(z+\frac{d}{2})^2} \right\}$$

$$= \frac{q}{4\pi\epsilon_0} \left\{ \frac{1}{z^2(1-\frac{d}{2z})^2} - \frac{1}{z^2(1+\frac{d}{2})^2} \right\}$$

$$= \frac{q}{4\pi\epsilon_0} \times \frac{1}{z^2} \left\{ \frac{1}{(1-\frac{d}{2z})^2} - \frac{1}{(1+\frac{d}{2})^2} \right\}$$

$$= \frac{1}{4\pi\epsilon_0} \times \frac{q}{z^2} \left\{ (1-\frac{d}{2z})^{-2} - (1+\frac{d}{2})^{-2} \right\}$$

Topic Name : Day : Time ; Date : / /

Here $\frac{d}{2z} << 1$.

So, $E = \frac{1}{4\pi\epsilon_0} \times \frac{q}{z^2}\left[1-(-2)\times\frac{d}{2z} - \left\{1+(-2)\frac{d}{2z}\right\}\right]$

$E = \frac{1}{4\pi\epsilon_0} \times \frac{q}{z^2}\left[\cancel{1}+\frac{2d}{2z} - \cancel{1}+\frac{2d}{2z}\right]$

$E = \frac{1}{4\pi\epsilon_0} \times \frac{q}{z^2}\left[2\times\frac{d}{z}\right]$

$$\boxed{E = \frac{1}{4\pi\epsilon_0} \times \frac{q}{z^2} \times \frac{2d}{z}}$$

$E = \frac{2qd}{4\pi\epsilon_0 z^3}$

$E = \frac{qd}{2\pi\epsilon_0 z^3}$

So, $\boxed{E = \frac{qd}{2\pi\epsilon_0 z^3}}$

$$\boxed{E = \frac{P}{2\pi\epsilon_0 z^3}}$$

if $d << z$ then

$\frac{d}{2z} << 1$

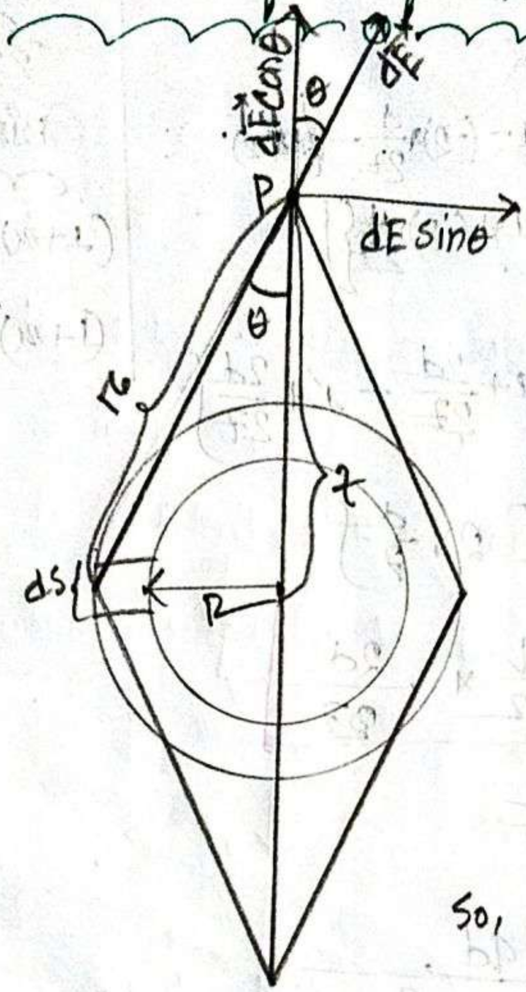
Binomal theoreme —

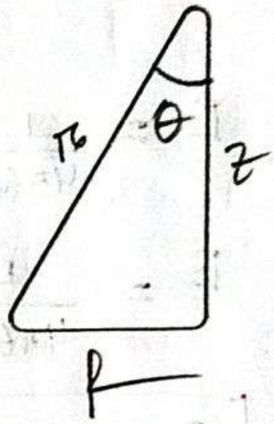
$(1+x)^n = 1 + \frac{nx}{1!} + \frac{n(n-1)x^2}{2!} + \cdots (x^2 < 1)$

$(1+x)^n \approx 1+nx$ for $x << 1$

Topic Name : ______ Day : ______ Time : Date : / /

Electric field for A ring:—





$$\cos\theta = \frac{z}{r_c}$$

$$r_c = \sqrt{R^2+z^2}$$

$$\text{So, } \cos\theta = \frac{z}{\sqrt{R^2+z^2}}$$

$$dE = \frac{1}{4\pi\epsilon_0} \times \frac{dq}{r_c^2}$$

$$= \frac{1}{4\pi\epsilon_0} \times \frac{\lambda ds}{r_c^2}$$

$$= \frac{1}{4\pi\epsilon_0} \times \frac{\lambda ds}{\left(\sqrt{z^2+R^2}\right)^2}$$

for linear charge density $\lambda = \frac{dq}{ds}$

Topic Name : Day : Time : Date : / /

So, $dE\cos\theta = \frac{1}{4\pi\epsilon_0} \times \frac{\lambda ds}{(\sqrt{R^2+z^2})^2} \times \frac{z}{\sqrt{R^2+z^2}}$

$$E_z = \int_0^{2\pi R} dE\cos\theta = \int_0^{2\pi R} \frac{1}{4\pi\epsilon_0} \times \frac{\lambda ds}{(\sqrt{R^2+z^2})^2} \times \frac{z}{\sqrt{R^2+z^2}}$$

$$= \frac{1}{4\pi\epsilon_0} \times \frac{\lambda z}{\sqrt{R^2+z^2}} \times \frac{1}{(R^2+z^2)^{\frac{2}{2}}} \int_0^{2\pi R} ds$$

$$= \frac{\lambda z}{4\pi\epsilon_0 (R^2+z^2)^{\frac{3}{2}}} \int_0^{2\pi R} ds$$

$$= \frac{\lambda z}{4\pi\epsilon_0 \; (R^2+z^2)^{\frac{3}{2}}} [s]_0^{2\pi R}$$

$$= \frac{\lambda z \times 2\pi R}{4\pi\epsilon_0 (R^2+z^2)^{\frac{3}{2}}}$$

$$\boxed{E_z = \frac{qz}{4\pi\epsilon_0 (R^2+z^2)^{\frac{3}{2}}}}$$

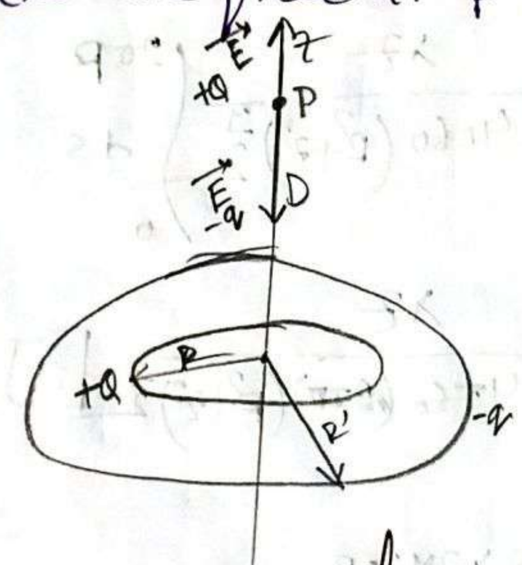
$\lambda = \frac{dq}{ds}$

$dq = \lambda \times ds$

$q = \lambda \times 2\pi R$

Problem 30:-

⇒ Figure shows two concentric rings, of radii R and $R' = 3R$, that lie on the same plane. point P lies on the central z axis, at distance $D = 2R$ from the center of the rings. The smaller ring has uniformly distributed charge +q. In terms of q, what is the uniformly distributed charge on the longer ring if the net electric field at P is zero.



⇒ According to the statement of the problem,

$$E_{+q} - E_{-q} = 0$$

$$E_{+q} = E_{-q} \quad \text{——— (1)}$$

$D = 2R$

1st circle $R = R$

2nd " $R' = 3R$

we know,

$$E = \frac{qz}{4\pi\epsilon_0 (z^2 + R^2)^{\frac{3}{2}}}$$

Topic Name : ______ Day : ______ Time : Date : / /

Substitude the Value of E in equation (1)

$$\Rightarrow \frac{QD}{4\pi\epsilon_0 (D^2+R^2)^{\frac{3}{2}}} = \frac{qD}{4\pi\epsilon_0 (D^2+R'^2)^{\frac{3}{2}}}$$

$$\Rightarrow \frac{Q}{(D^2+R^2)^{\frac{3}{2}}} = \frac{q}{(D^2+(3R)^2)^{\frac{3}{2}}}$$

$$\Rightarrow \frac{Q}{q} = \frac{(D^2+R^2)^{\frac{3}{2}}}{(D^2+9R^2)^{\frac{3}{2}}}$$

$$\Rightarrow \frac{Q}{q} = \frac{(4R^2+R^2)^{\frac{3}{2}}}{(4R^2+9R^2)^{\frac{3}{2}}}$$

$$\Rightarrow \frac{Q}{q} = \left(\frac{5R^2}{13R^2}\right)^{\frac{3}{2}}$$

$$\Rightarrow \frac{Q}{q} = \left(\frac{5}{13}\right)^{\frac{3}{2}}$$

$$\Rightarrow \frac{Q}{q} = 0.23$$

$$\Rightarrow \frac{q}{Q} = \frac{1}{0.23}$$

$$\Rightarrow q = 4.19\,Q$$

$$q = -4.19\,Q$$

Ch: 23

GAUSS' LAW

Topic Name : Day : Time : Date : / /

Q:- What is electric Flux?—

⇒ Electric flux is the measure of the total numbers of electric field that pass through a particular Surface.

Electric Flux for uniform $\vec{E}$ and through a flat surface (open plane):-

⇒ The electric flux ϕ through the flat surface equals the scalar product of the electric field $\vec{E}$ and the area vector $\vec{A}$

$$\boxed{\phi = \vec{E} \cdot \vec{A}}$$

$$\boxed{\phi = EA\cos\theta}$$

$$\boxed{\begin{aligned} \phi_{max} &= EA\cos 0^\circ \\ \phi_{mini} &= EA\cos 90^\circ \end{aligned}}$$

~~cgs unit~~ / SI Unit $= N\text{-}m^2/C$

P.T.O

Topic Name : ______ Day : ______ Time : Date : / /

Q: Write down the condition of electric flux:- ?

1. electric flux maximum ; when $\theta = 0°$
2. electric flux minimum ; when $\theta = 90°$

electric flux for nonuniform $\vec{E}$ And through a curved surface:-

⇒ When the surface is carved and field is nonuniform, we calculate the flux by dividing the surface into small patches dA, so that each differential area is assentially flat and field is uniform over each patch.

$$\boxed{\Phi = \int \vec{E} \cdot d\vec{A}}$$

P.T.O

Topic Name : Time : Date : / /

Electric Flux through a closed Surface:

⇒ The net flux through a closed surface is given by

$$\phi = \oint \vec{E} \cdot \vec{dA}$$

$$\boxed{\phi = \int \vec{E} \cdot \vec{dA} + \int \vec{E} \cdot \vec{dA} + \int \vec{E} \cdot \vec{dA}}$$

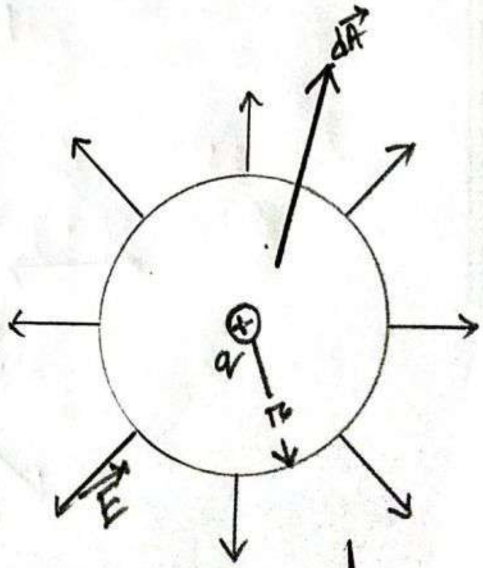
Gauss' Law:

⇒ Gauss's Law states that the total electric flux(ϕ) through any closed surface, known as gaussain surface is proportional to the total (net) electric charge(q) inside the surface.

$$\boxed{\phi = \frac{q}{\epsilon_0}}$$

$$\boxed{\phi \epsilon_0 = q}$$

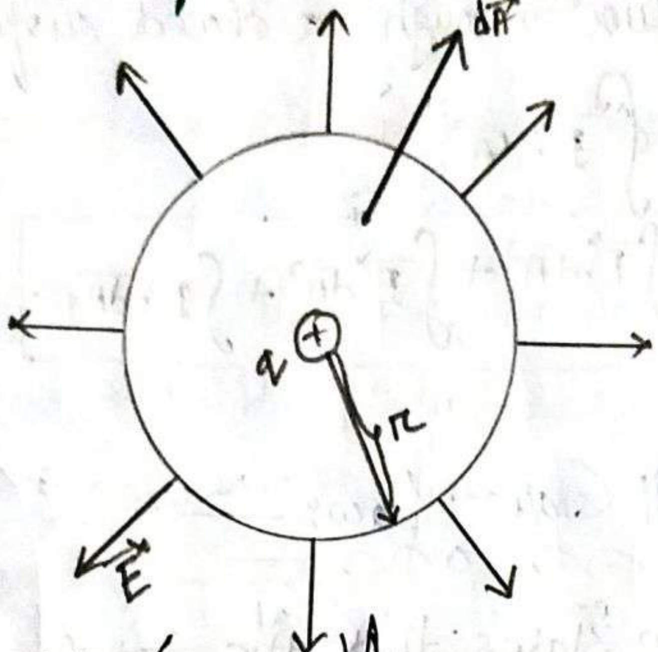
$$\boxed{\epsilon_0 \oint \vec{E} \cdot \vec{dA} = q}$$



Gaussain surface

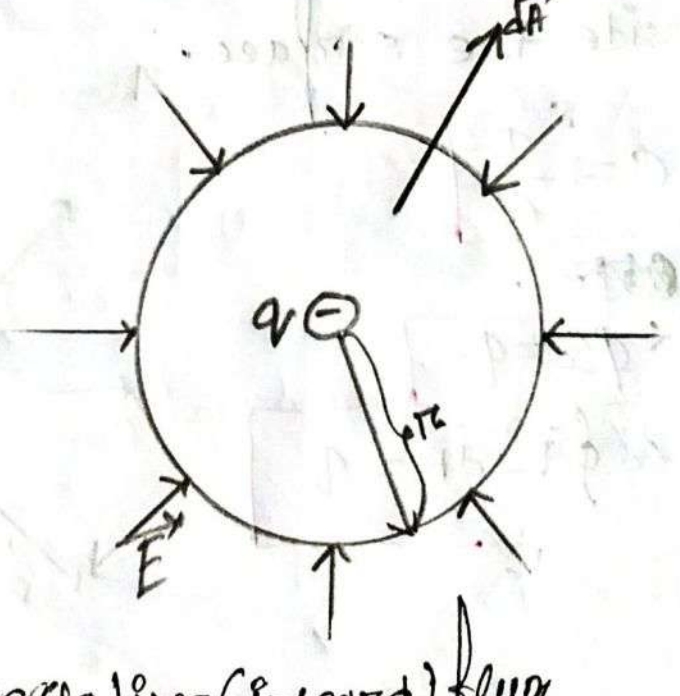
Topic Name : ______ Day : ______ Time : Date : / /

☒ Gaussian surface around positive charge:-



Positive (outward) flux

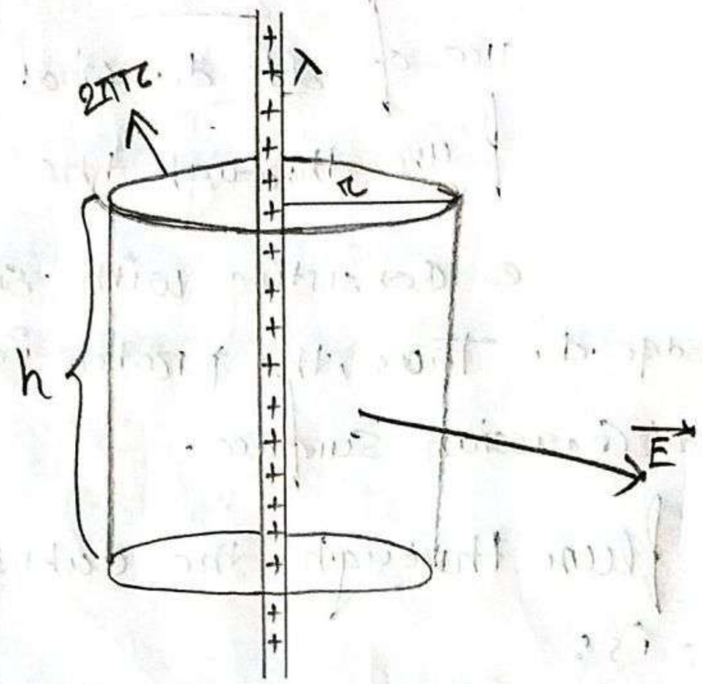
☒ Gaussian surface around negative charge:-



negative (inward) flux

Topic Name: ______ Day: ______ Time: Date: / /

Applying Gauss' Law: ~~for~~ Cylindrical Symmetry:-



⇒ An infinity long cylindical plastic rod with a uniform charge density $= \lambda$

radius $= r$ | $A = 2\pi r h$
length $= h$

using Gauss' law, we can write.

$$\epsilon_0 \Phi = q = \lambda h$$

$$\epsilon_0 \Phi = \lambda h$$

$$\epsilon_0 E A = \lambda h$$

$$\epsilon_0 E (2\pi r h) = \lambda h$$

$$\boxed{E = \frac{\lambda}{\epsilon_0 2\pi r}}$$

Where,
λ = the liner charge density.

Topic Name : Day : Time : Date : / /

Problem: 5:

⇒ A proton is a distance $\frac{d}{2}$ directly above the center of a square of side d. What is the magnitude of the electric flux through the square?

⇒ Consider a cube consisting with six square faces; each of edge d. Thus, the proton is enclosed by the Cubical Gaussian Surface.

⇒ Electric flux through the cubical Gaussian Surface is:

$$\phi = \frac{q}{\varepsilon_0} = \frac{e^+}{\varepsilon_0} = \frac{1.6\times10^{-19}}{8.854\times10^{-12}}$$

$$= 180\times10^{-9} \ N.m^2/C$$

So, Electric flux through a face of a cable is,

$$\phi_s = \frac{180\times10^{-9}}{6} = 30\times10^{-9} \ \frac{Nm^2}{C}$$

Ans:

Topic Name : Day : Time : Date : / /

Problem: 25

⇒ An infinite line of charge produces a field of magnitude 4.5×10^4 N/C. at distance 2m. Find the linear charge density.?

⇒ is given:—

$$E = 4.5\times10^4 N/C$$

$$r = 2m$$

$$\epsilon_0 = 8.85\times10^{-12}$$

we know:—

$$E = \frac{\lambda}{2\pi\epsilon_0 r}$$

$$\lambda = E\times2\pi\epsilon_0\times r$$

$$= 4.5\times10^4\times2\times\pi\times8.85\times10^{-12}$$

$$= 2.5\times10^{-6} c/m$$

Topic Name: Day: Time: Date: / /

Ch no 24

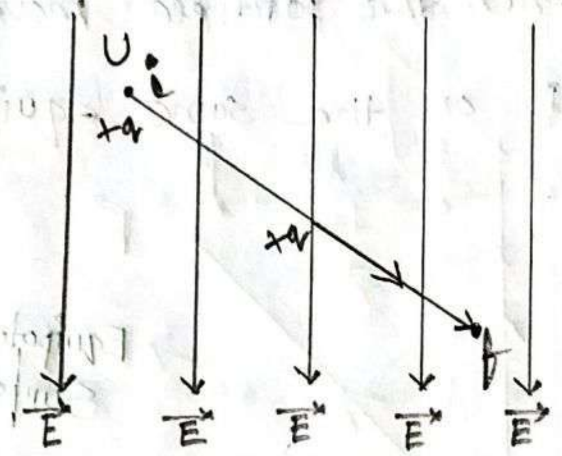
Electric Potential

Topic Name : ______ Day : ______ Time : Date : / /

Electric potential:

⇒ The potential energy (U) per unit charge (q) at a point in a electric field ($\vec{E}$) is called the electric potential.

$$\boxed{V = \frac{U}{q}}$$



the difference in potential energy per unit charge between the two points:—

$$\Delta V = V_f - V_i$$

$$\boxed{\Delta V = \frac{U_f}{q} - \frac{U_i}{q}}$$

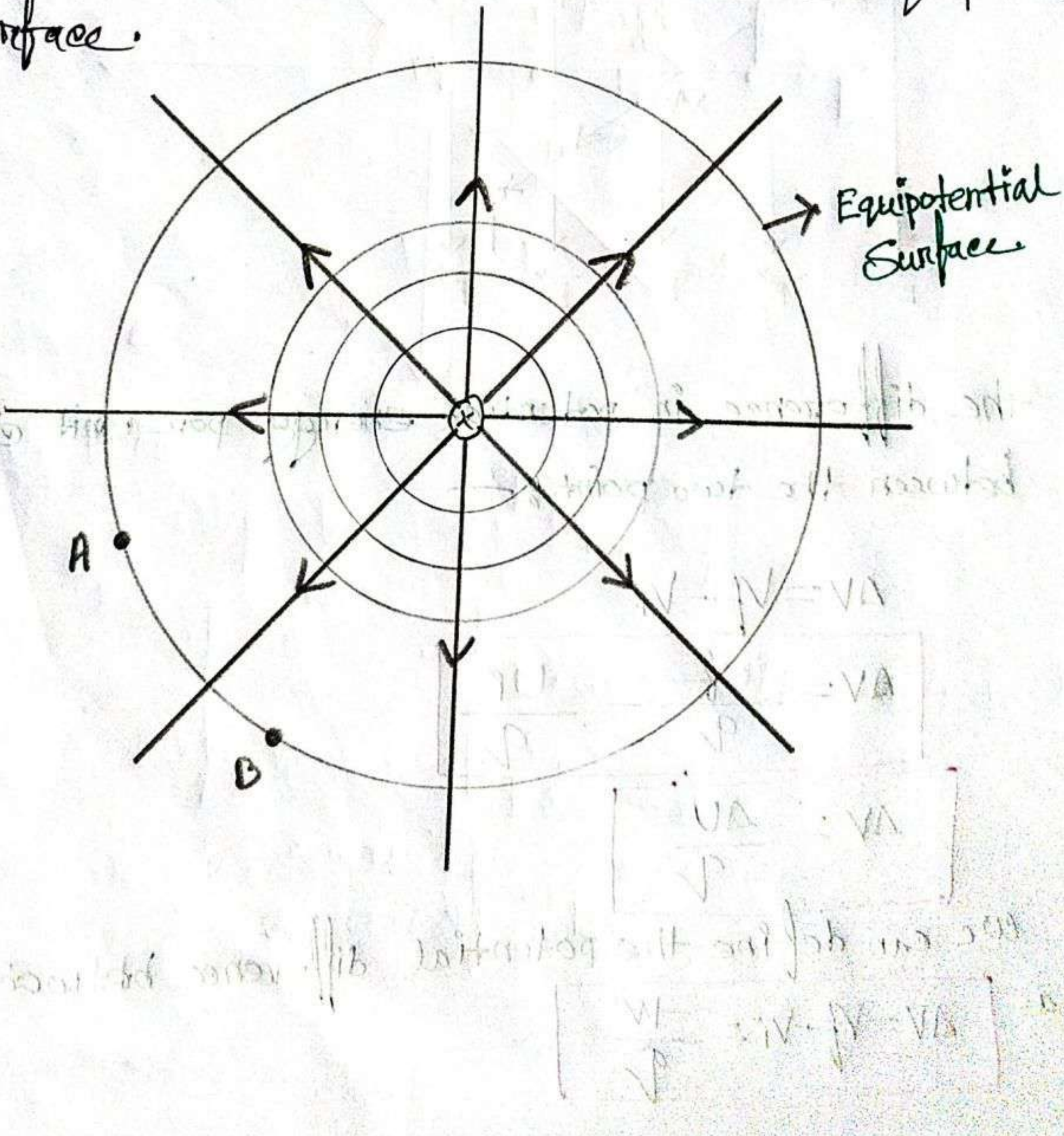
$$\boxed{\Delta V = \frac{\Delta U}{q}}$$

We can define the potential difference between points i and f as $\boxed{\Delta V = V_f - V_i = \frac{-W}{q}}$

Topic Name : ______________ Day : ______ Time : Date : / /

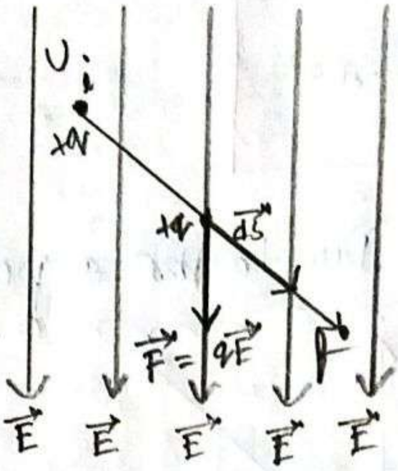
Equipotential Surfaces:—

⇒ Adjacent points that have the same electric potential form an equipotential surface, which can be either an imaginary surface or real, physical surface. No net work W is done on a charged particle by an electric field when the particle moves between two points. A and B on the same equipotential surface.



Topic Name : Day : Time : Date : / /

Calculating the potential from the fields:



We know,

$$dW = \vec{F} \cdot \vec{ds}$$

$$\boxed{dW = q\vec{E} \cdot \vec{ds}}$$

The total work W done on the particle by the field as the particle moves from poin i to point f,

$$W = \int_i^f dW = \int_i^f q\vec{E} \cdot \vec{ds} = q\int_i^f \vec{E} \cdot \vec{ds}$$

So, the work done by electrostatic force in terms of potential difference:

$$\Delta V = V_f - V_i = -\frac{W}{q}$$

$$V_f - V_i = \frac{-q\int_i^f \vec{E} \cdot \vec{ds}}{q}$$

P.T.O

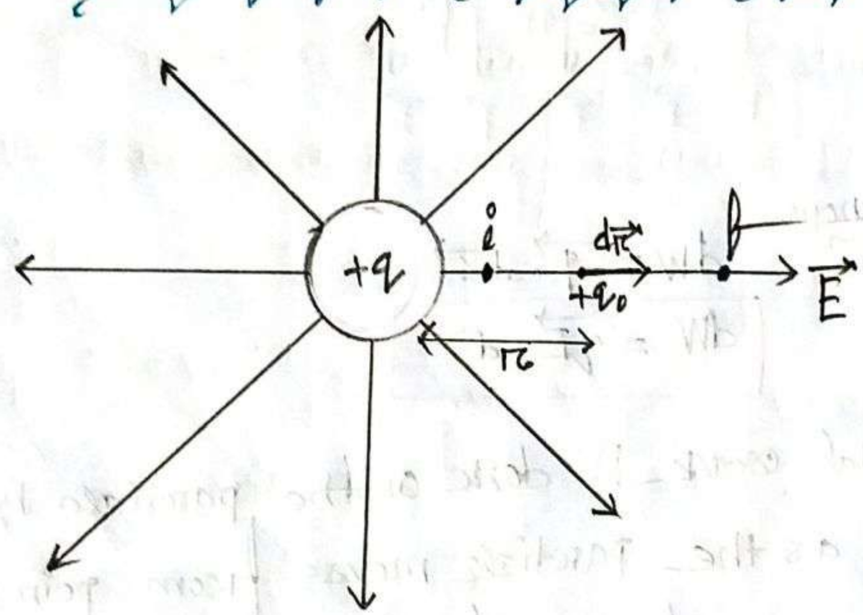
Topic Name: Day: lesson: 4

Time: Date: / /

$$V_f - V_i = -\int_i^f \vec{E}\cdot\vec{ds}$$

$$\boxed{V_f - V_i = -\int_i^f \vec{E}\cdot\vec{ds}}$$

potential Due to a point charge:-



We know,

$$V_f - V_i = -\int_i^f \vec{E}\cdot\vec{ds}$$

$$V_f - V_i = -\int_i^f \vec{E}\cdot\vec{dr} \quad [\text{here } \vec{ds} = \vec{dr}]$$

$$V_f - V_i = -\int_i^f E\cos 0^\circ \cdot dr$$

$$= -\int_i^f E\,dr$$

Topic Name : Day : Time : Date : / /

$$V_f - V_i = -\frac{q}{4\pi\epsilon_0}\int_i^f \frac{1}{r^2}\,dr$$

$$V_f - V_i = -\frac{q}{4\pi\epsilon_0}\left(-\left|\frac{1}{r}\right|_{r=r_i}^{r=r_f}\right)$$

$$\boxed{V_f - V_i = \frac{q}{4\pi\epsilon_0}\left[\frac{1}{r_f}-\frac{1}{r_i}\right]}$$

if $r_f = \alpha$ then $V_f = 0$ and $\frac{1}{r_f} = 0$

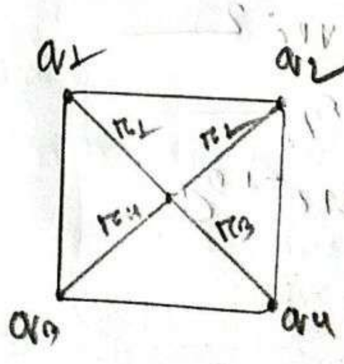
So,

$$0 - V_i = \frac{q}{4\pi\epsilon_0}\left[0-\frac{1}{r_i}\right]$$

$$V_i = \frac{q}{4\pi\epsilon_0}\left[\frac{1}{r_i}\right]$$

$$\therefore \boxed{V = \frac{1}{4\pi\epsilon_0}\times\frac{q}{r}}$$

Potential Due to the Group of point Charges:—



Topic Name : Day : Time : Date : / /

$$V=\sum_{i=1}^{n} V_i = \frac{1}{4\pi\epsilon_0}\sum_{i=1}^{n}\frac{q_i}{r_i}$$

$$\boxed{V=\sum_{i=1}^{4} V_i = \frac{1}{4\pi\epsilon_0}\left[\frac{q_1}{r_1}+\frac{q_2}{r_2}+\frac{q_3}{r_3}+\frac{q_4}{r_4}\right]}$$

Q: No 4:-

⇒ Two large, parallel conducting plates are 12 cm apart and have a charge of equal magnitude and opposite sign on their facing surfaces. An electric force of $3.9\times10^{-15}N$ acts on an electric placed anywhere between the two plates.

(a) Find the electric field at the position of the electron.

(b) What is the potential difference between the plates?

⇒ is given,

$d = 12cm = 12\times10^{-2}$

$F = 3.9\times10^{-15}N$

$e^- = q = 1.6\times10^{-19}C$

Topic Name : Day : Time : Date : / /

(a)

We know,

$$\boxed{F = qE}$$

$$E = \frac{F}{q}$$

$$= \frac{3.9 \times 10^{-15}}{1.6 \times 10^{-19}}$$

$$= 2.4375 \times 10^{4} \text{ N/C}$$

(b)

We know:—

$$\boxed{|\Delta V| = Ed}$$

$$= 2.4375 \times 10^{4} \times 12 \times 10^{-2}$$

$$= 2925 \text{ Volt}$$

Problem: 6:—

When an electron moves from A to B along an electric field line in the adjacent figure, the electric field does 3.94×10^{-19} J of work on it. What are the electric potential differences

(a) $V_B - V_A$

(b) $V_C - V_A$

(c) $V_C - V_B$

Topic Name : Day : Time : Date : / /

⇒ is given,

$W = 3.94 \times 10^{-19} J$

$q = 1.6 \times 10^{-19} C$

(a)

$V_B - V_A = \frac{-W}{q}$

$= \frac{-3.94 \times 10^{-19}}{-1.6 \times 10^{-19}}$ Volt

$= 2.4625$ Volt

(b)

$V_C - V_A = \frac{-W}{q}$

$= \frac{-3.94 \times 10^{-19}}{-1.6 \times 10^{-19}}$ Volt

$= 2.4625$ Volt

(c)

$V_C - V_B = 0$ Volt

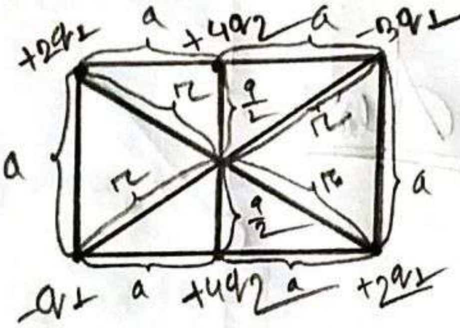
Because points B and C are the same equipotential surface.

Topic Name : Day : Time : Date : / /

Problem: 16:—

⇒ Figure shows a rectangular array of charged particles fixed in place, with distance $a = 30$ cm. and the charge shown as integer multiples of $q_1 = 3.4$ pC and $q_2 = 6$ pC. with $V = 0$ at infinity, what is the net electric potential at the rectangle's center?

⇒



is given:—

$a = 30$ cm $= 0.30$ m

$q_1 = 3.40 \times$ pC $= 3.40 \times 10^{-12}$ C

$q_2 = 6$ pC $= 6.0 \times 10^{-12}$ C

We know:—

The net potential at the ~~rec~~ rectangle center is:—

$$V = \frac{1}{4\pi\epsilon_0}\left[\frac{2q_1}{r} + \frac{-3q_1}{r} + \frac{2q_1}{r} + \frac{-q_1}{r}\right] + \frac{1}{4\pi\epsilon_0}\left(\frac{4q_2}{\frac{a}{2}} + \frac{4q_2}{\frac{a}{2}}\right)$$

$$= \frac{1}{4\pi\epsilon_0}\left(\frac{4q_1}{2r} - \frac{4q_1}{2r}\right) + \frac{1}{4\pi\epsilon_0}\left(\frac{4q_2}{\frac{0.30}{2}} + \frac{4q_2}{\frac{0.30}{2}}\right)$$

P.T.O

Topic Name : Day : Time : Date : / /

$$V = 0 + \frac{1}{4\pi\epsilon_0}\left[\frac{4q_2}{0.105} + \frac{4q_2}{0.105}\right]$$

$$V = \frac{1}{4\pi\epsilon_0}\left[\frac{8q_2}{0.105}\right]$$

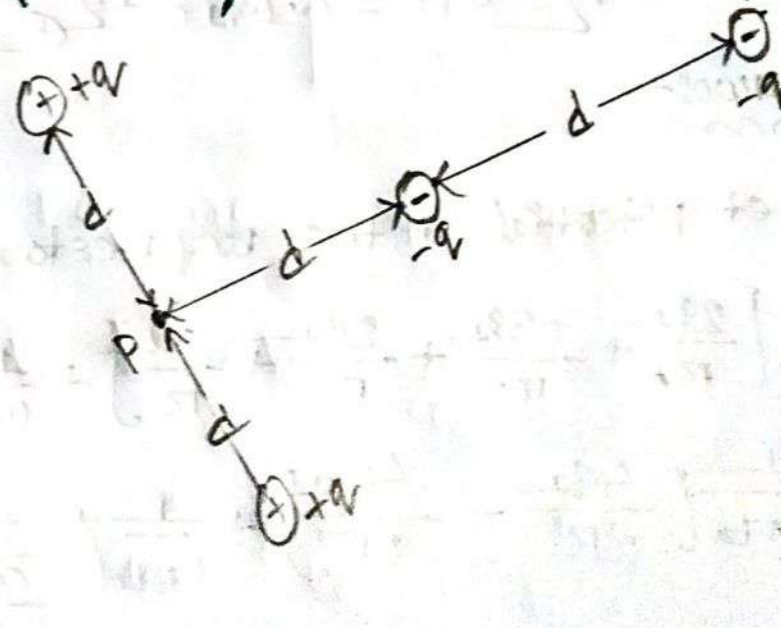
$$= 9\times10^9 \times \frac{8\times 6\times10^{-12}}{0.105}$$

$$= 2.215 \text{ Volt}$$

Ans:—

Problem: 17:—

⇒ In the adjacent figure, what is the net electric potential at point p due to the four particles if $V=0$ at infinity, $q = 5fC$, and $d = 4$ cm ?



⇒ is given:—

$q = 5fC = 5 \times 10^{-15} C$

$d = 4cm = 0.04m$

We know:—

The net electric potential at the point p is:—

$$V = \frac{1}{4\pi\epsilon_0}\left(\frac{q}{d} + \frac{q}{d} + \frac{-q}{d} + \frac{-q}{2d}\right)$$

$$= \frac{1}{4\pi\epsilon_0} \times \frac{q}{d}\left(1 + 1 - 1 - \frac{1}{2}\right)$$

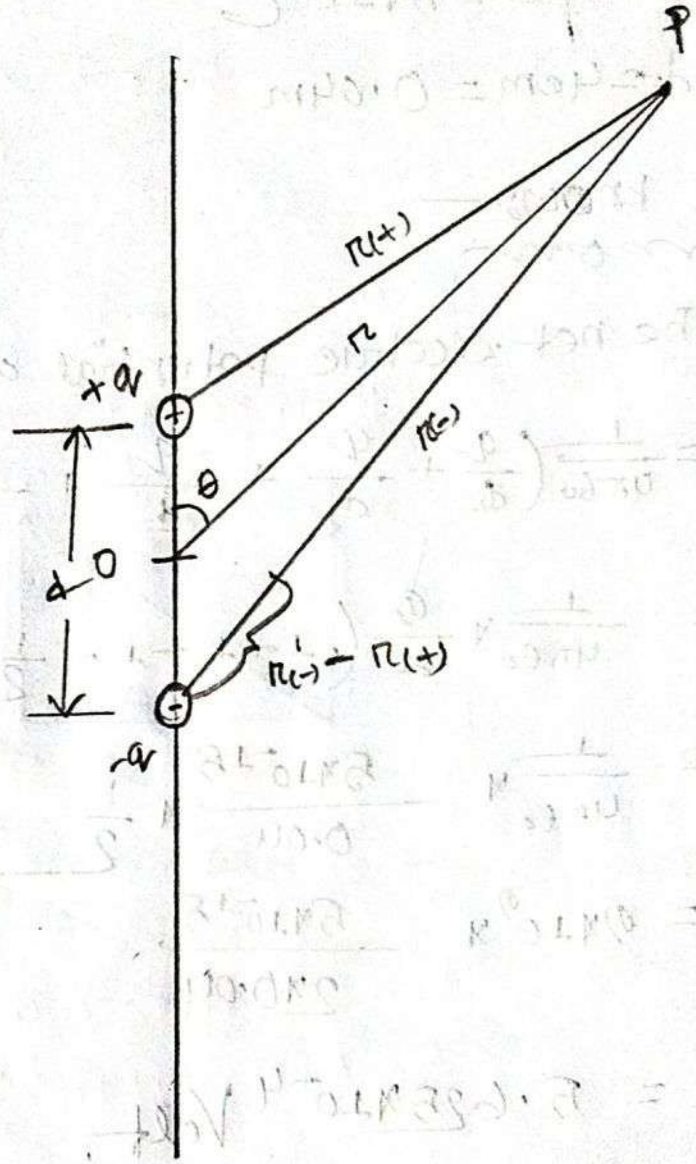
$$= \frac{1}{4\pi\epsilon_0} \times \frac{5 \times 10^{-15}}{0.04} \times \frac{1}{2}$$

$$= 9 \times 10^9 \times \frac{5 \times 10^{-15}}{2 \times 0.04}$$

$$= 5.625 \times 10^{-4} \text{ Volt.}$$

Topic Name : ______ Day : ______ Time : Date : / /

#potential Due to an Electric Dipole:-



the net potential at P is:-

$$V = \sum_{i=1}^{2} V_i = V_{(+)} + V_{(-)} = \frac{1}{4\pi\epsilon_0}\left(\frac{q}{r_{(+)}} + \frac{-q}{r_{(-)}}\right)$$

$$V = \frac{q}{4\pi\epsilon_0}\left(\frac{1}{r_{(+)}} + \frac{-1}{r_{(-)}}\right)$$

Topic Name : Day : Time : Date : / /

$$V = \frac{q}{4\pi\epsilon_0}\left(\frac{r_{(-)} - r_{(+)}}{r_{(+)} \times r_{(-)}}\right) \quad \text{——— (iii)}$$

for dipole $d << r_0$.

So, we can write,

$r_{0(-)} - r_{0(+)} = d\cos\theta$. and ——— (i)

$r_{(-)}\, r_{0(+)} = r_0^2$ ——— (ii)

Substing these value in number (iii) ⇒

$$V = \frac{q}{4\pi\epsilon_0}\left(\frac{d\cos\theta}{r^2}\right)$$

$$\boxed{V = \frac{1}{4\pi\epsilon_0}\left(\frac{p\cos\theta}{r^2}\right)}$$ where $p = qd$

The vector $\vec{p}$ is directed along the dipole axis from the negative to the positive charge.

$\vec{p}$ = is the magnitude of electric dipole moment

Topic Name :

Day : Lesson: 06

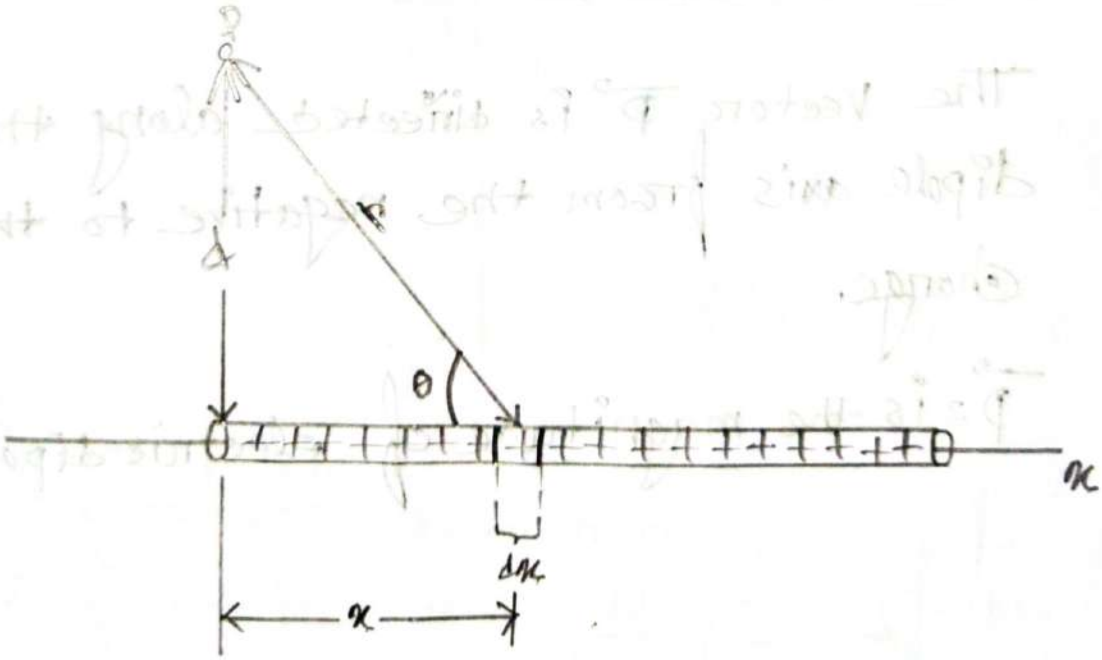
Time : Date : / /

Potential due to a line of Charge:-

Let, A thin nonconducting rod which have a positive charge.

L = Length of the rod
λ = Linear charge density.
V = electric potential
d = distance

Now we consider a differential element = dx
This element of the rod has a differential charge, $\boxed{dq = \lambda dx}$ - - - - - (i)



$$r = \sqrt{d^2 + x^2} \quad \text{- - - - - (ii)}$$

$$\tan\theta = \frac{d}{x}$$

$$x = d\cot\theta$$

$$dx = -d\,\mathrm{cosec}^2\theta \cdot d\theta$$

Topic Name : Day : Time : Date : / /

Let, the element as a point charge,

So, $dv = \frac{1}{4\pi\epsilon_0} \cdot \frac{dq}{r}$

Substitude the Value of dq and r =>

$$dv = \frac{1}{4\pi\epsilon_0} \times \frac{\lambda dx}{\sqrt{d^2+x^2}}$$

$$V = \int dv = \int_{x=0}^{x=L} \frac{1}{4\pi\epsilon_0} \times \frac{\lambda dx}{(d^2+x^2)^{\frac{1}{2}}}$$

$$= \frac{\lambda}{4\pi\epsilon_0} \int_0^L \frac{dx}{(d^2+x^2)^{\frac{1}{2}}}$$

$$= \frac{\lambda}{4\pi\epsilon_0} \int_0^L \frac{-d\,\text{cosec}^2\theta \cdot d\theta}{(d^2+d^2\cot^2\theta)^{1/2}}$$

$$= \frac{1\lambda}{4\pi\epsilon_0} \int_0^L -\text{cosec}\,\theta\, d\theta$$

$$= \frac{1\lambda}{4\pi\epsilon_0} \times \int_0^L [\ln|\text{cosec}\,\theta + \cot\theta|]$$

$$= \frac{1\lambda}{4\pi\epsilon_0} \int_0^L \ln\left|\frac{\sqrt{d^2+x^2}}{d} + \frac{x}{d}\right|$$

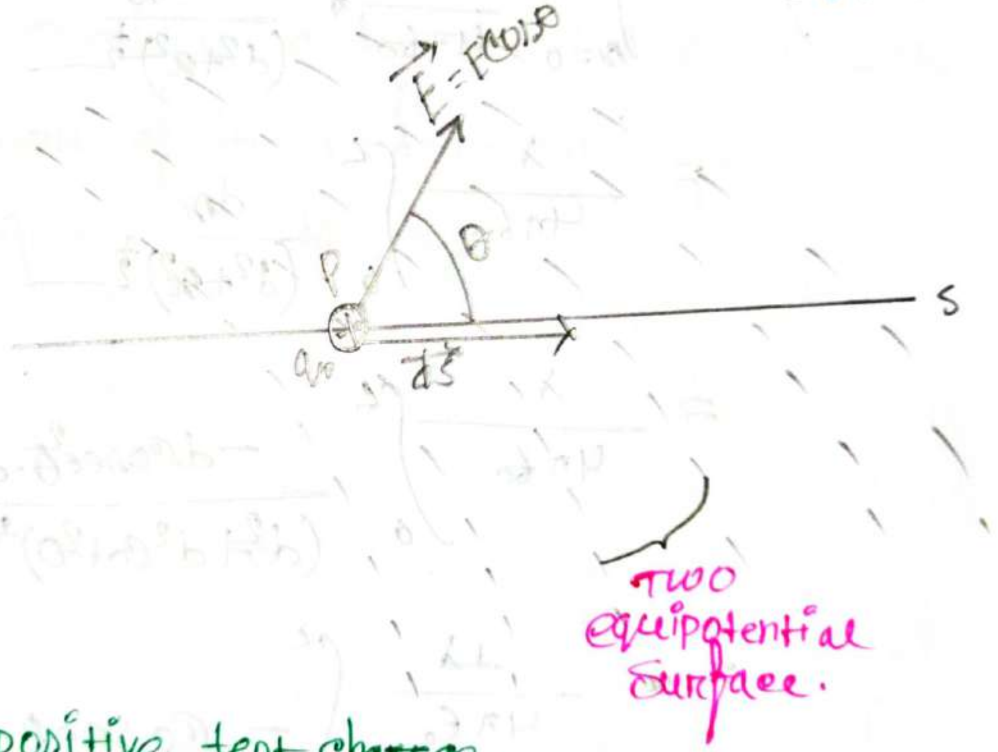
$$= \frac{1\lambda}{4\pi\epsilon_0} \left[\ln\left|\frac{\sqrt{d^2+L^2}}{d} + \frac{L}{d}\right| - \ln\frac{\sqrt{d^2-0^2}}{d} + \frac{0}{d}\right]$$

Topic Name : Day : Time : Date : / /

$$\boxed{V = \frac{\lambda}{4\pi\epsilon_0} \ln\left|\frac{\sqrt{d^2+L^2}+L}{d}\right|}$$

V = The total potential produced by the rod at point p

Calculating the electric field from the Electric Potential:-



q_0 = positive test charge..

$\vec{ds}$ = displacement from equipotential surface to adjacent Surface.

dv = potential difference

dw = differential work.

$\vec{E}$ = Electric field.

Topic Name : Day : Time : Date : / /

Work done by electric potential:-

So, $dW = -q_0 dV$ —— (I) | or, we know, $dV = \frac{-dW}{q_0}$

Again:-

Work done by electric field:-

$dW = F \times ds \times \cos\theta$

$= Eq_0 \times ds \times \cos\theta$ —— (II)

we know, $E = \frac{F}{q_0}$

$F = Eq_0$

Now, we can writes:-

$dw = dw$

$-q_0 dV = q_0 E \times ds \times \cos\theta$

$E \times \cos\theta = \frac{-q_0 dV}{q_0 ds}$

$E \times \cos\theta = -\frac{dV}{ds}$

Since $E\cos\theta$ is the component of $\vec{E}$ in the direction of $d\vec{s}$,

So, $\boxed{E_s = -\frac{\partial V}{\partial s}}$

So, x, y, and z axes components of $\vec{E}$ at any point are,

$$\boxed{E_x = -\frac{\partial V}{\partial x};\quad E_y = -\frac{\partial V}{\partial y};\quad E_z = \frac{-\partial V}{\partial z}}$$

Topic Name : Day : Lesson 86

Time : Date : / /

Problem 22 :—

⇒ The ammonia molecule NH_3 has a permanent electric dipole moment equal to 1.47D, where 1D = 1 debye unit = 3.34×10^{-30} C.m Calculate the electric potential due to an ammonia molecule at a point ~~at a point~~ 52.0 nm away along the axis of the dipole. (Set V=0 at infinity)

⇒ is given,

$$P = 1.47D = 1.47\times3.34\times10^{-30}\ C.m$$

$$r = 52nm = 10^{-9}\times52$$

We know:—

$$V = \frac{1}{4\pi\epsilon_0}\times\frac{q}{r}$$

$$= \frac{1}{4\pi\epsilon_0}\times\frac{q}{r}\times\frac{r}{r}$$

$$= \frac{1}{4\pi\epsilon_0}\times\frac{qr}{r^2}$$

[Dipole moment $P = qd$]

$$= \frac{1}{4\pi\epsilon_0}\times\frac{P}{r^2}$$

$$= 9\times10^9\times\frac{1.47\times3.34\times10^{-30}}{(52\times10^{-9})^2}$$

$$= 1.63\times10^{-5}\ \text{Volt}.$$

Topic Name : Day : Time : Date : / /

Problem 36:—

⇒ The electric potential V in the space between flat Parallel plates 1 and 2 is given (in Volts) by $V=1500x^2$, Where x (in meter) is the perpendicular distance from Plate 1. at $x=1.3$ cm,

(a) What is the magnitude of the electric field?

(b) Is the field directed toward, away from plate 1?

⇒ is given:—

$V=1500x^2$

$x=1.3\text{ cm}=1.3\times10^{-2}\text{ m}$

(a)

We know:—

$$E_x=\frac{-dV}{dx}$$

$$=\frac{-d}{dx}\times1500x^2$$

$$=-2\times1500x$$

$$=-3000x$$

$$=-3000\times1.3\times10^{-2}$$

$$=-39\text{ Vm}^{-1}$$

So, $\vec{E}=E_x\hat{i}=39(-\hat{i})$

Magnitude of $\vec{E}$ is:—

$|\vec{E}|=\sqrt{(-39)^2}=39$

$E=39\text{ Vm}^{-1}$

Ans:—

P.T.O

Topic Name : ______ Day : ______
Time : Date : / /

(b)

$\Rightarrow$ The direction of electric field toward plate 1.

because $\vec{E} = 30(-\hat{i})\ Vm^{-1}$

Problem: 378

$\Rightarrow$ What is the magnitude of the electric field at the point $(3\hat{i} - 2\hat{j} + 4\hat{k})m$ if the electric potential in the region is given by $V = 2xyz^2$, where V is the Volts and coordinates x, y, and z are in meters?

$\Rightarrow$ is given:

$V = 2xyz^2$

$\langle x, y, z \rangle = \langle 3, -2, 4 \rangle$

we know:

$E_x = -\frac{\partial V}{\partial x} = -\frac{\partial}{\partial x} 2xyz^2 = -2yz^2$

$= -2 \times -2 \times (4)^2$

$= 64\ Vm^{-1}$

$E_y = -\frac{\partial V}{\partial y} = -\frac{\partial}{\partial y} 2xyz^2 = -2xz^2$

$= -2 \times 3 \times 4^2$

$= -96\ Vm^{-1}$

Topic Name : ______ Day : ______ Time : Date : / /

$$E_z = -\frac{\partial V}{\partial z} = -\frac{\partial}{\partial z} 2xyz^2 = -4xyz$$

$$= -4 \times 3 \times -2 \times 4$$

$$= 96 \ Vm^{-1}$$

$$\vec{E} = E_x\hat{i} + E_y\hat{j} + E_z\hat{k}$$

$$= 64\hat{i} - 96\hat{j} + 96\hat{k}$$

$$|\vec{E}| = \sqrt{(64)^2 + (-96)^2 + (96)^2}$$

$$= 150.00 \ Vm^{-1}$$

Topic Name : Day :

Time : Date : / /

Ch 25

Capacitance

Topic Name : Day : Time : Date : / /

Capacitor:-

⇒ A capacitor is an electronic component that stores electrical energy in the electric field, Consisting of two Conductive plates separated by a dielectric meterial.

Capacitance:-

⇒ A Capacitor Consists of two isolated Conductors or plates with Charge $+q$ and $-q$.

A Charge q and the potential difference V for the Capacitor are proportional to each other.

$$q \propto V$$

$$\boxed{q = CV}$$

C = Capacitance of capacitor.

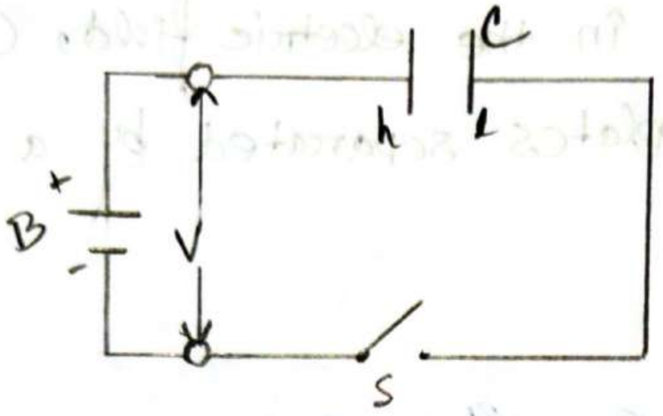
So, $\boxed{C = \frac{q}{V}}$

SI Unit CV^{-1}

Common name is Farad (F);

Topic Name : Day : Time : Date : / /

Charging a Capacitor:-



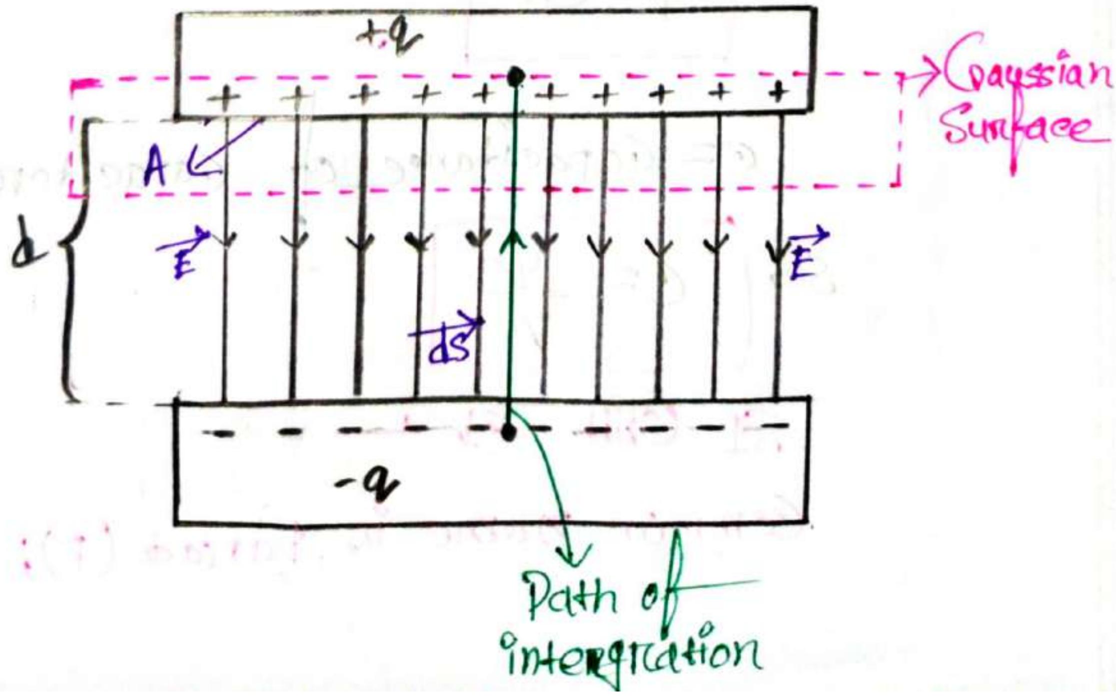
B = Battery

S = switch

h & l = plates of capacitor C.

Calculating the Capacitance:-

A parallel-plate Capacitor:-



Topic Name : Day : Time : Date : / /

A = Area of that part of the Gaussian surface through which there is a flux.

Applying Gauss' Law:—

We know:—

$$\phi = \frac{q}{\epsilon_0}$$

$$\phi \epsilon_0 = q$$

$$\epsilon_0 \oint \vec{E} \cdot d\vec{A} = q \quad [\phi = \oint \vec{E} \cdot d\vec{A}]$$

q = charge

$\oint \vec{E} \cdot d\vec{A}$ = net electric flux

$$\epsilon_0 \oint E \cdot dA \cdot \cos 0° = q$$ [Since $\vec{E}$ and $d\vec{A}$ are Parallel]

$$\boxed{\epsilon_0 E A = q}$$

The potential difference between the plates of a capacitor is related to the field $\vec{E}$.

$$V_f - V_i = -\int_i^f \vec{E} \cdot d\vec{s}$$

$$= -\int_i^f E \cos 180° \, ds$$

$$V = \int_-^+ E ds \quad [V = V_f - V_i]$$

$$V = E \int_{S=0}^{S=d} ds$$

$$\boxed{V = Ed}$$

Topic Name : Day : Time : Date : / /

we have:–

$$C = \frac{q}{V}$$

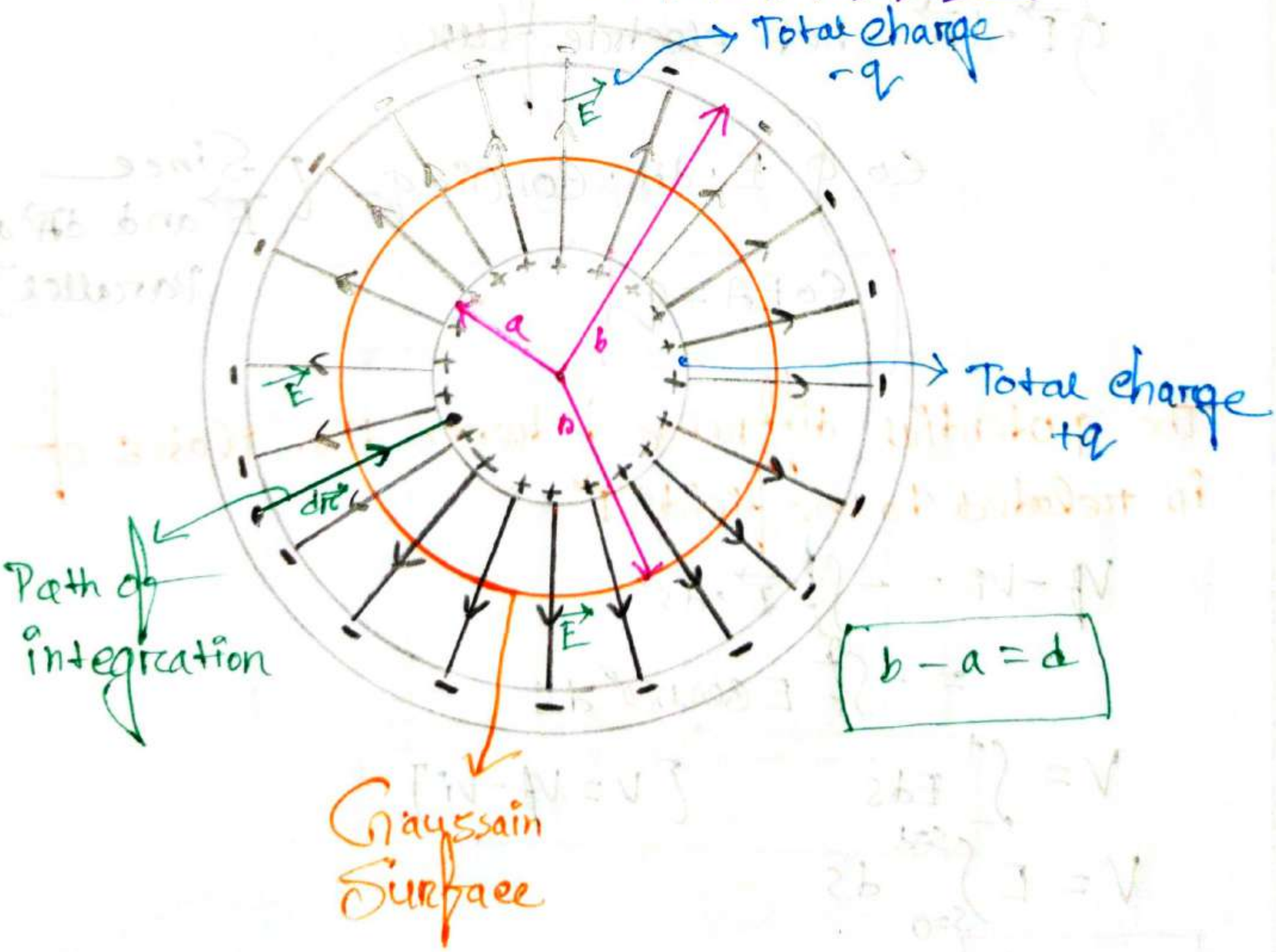
Substitute the value of V and q

$$C = \frac{q}{Ed}$$

$$C = \frac{\epsilon_0 EA}{Ed}$$

$$\boxed{C = \frac{\epsilon_0 A}{d}} \; \boxed{= \frac{\epsilon_0 (\pi r^2)}{d}} \rightarrow \text{for circular plates}$$

A Spherical Capacitor:–



$$\boxed{b - a = d}$$

Topic Name :

Day: Lesson 7 & 8

Time : Date : / /

r = sphere of radious

a & b = two concentric spherical shells of radii

Applying Gauss' Law:—

We can find:—

$$\epsilon_0 EA = q$$

$$\epsilon_0 E(4\pi r^2) = q$$

$$E = \frac{q}{4\pi\epsilon_0 r^2} \quad \text{--- (i)}$$

where, $A = 4\pi r^2$ is the area of the spherical Gaussian Surface.

We know:—

$$V = \int_-^+ E\,ds = -\int_-^+ E\,dr \quad [\text{Since } ds = -dr]$$

$$V = -\frac{q}{4\pi\epsilon_0}\int_{r=b}^{r=a}\frac{dr}{r^2}$$

$$= \frac{q}{4\pi\epsilon_0}\left(\left|\frac{1}{r}\right|_b^a\right)$$

$$= \frac{q}{4\pi\epsilon_0}\left(\frac{1}{a}-\frac{1}{b}\right)$$

$$= \frac{q}{4\pi\epsilon_0}\times\frac{b-a}{ab}$$

$$= \frac{q(b-a)}{4\pi\epsilon_0 ab} \quad \text{--- (ii)}$$

Substitude the Value of V in the equation of

$$C = \frac{q}{V}$$

$$= \frac{q}{\frac{q(b-a)}{4\pi\epsilon_0 \times ab}}$$

$$= q \times \frac{4\pi\epsilon_0 \times ab}{q(b-a)}$$

$$C = \frac{4\pi\epsilon_0 \times ab}{(b-a)}$$

$$\therefore \boxed{C = \frac{4\pi\epsilon_0 \times ab}{(b-a)}}$$

An isolated sphere:-

⇒ We can assaign a capacitance to a single isolated spherical conductor of radius R by assuming that ~~the~~ "missing plate" in a conducting sphere of infinite radius.

We know:-

$$C = \frac{4\pi\epsilon_0 \times ab}{b-a}$$

$$\Rightarrow C = \frac{\frac{4\pi\epsilon_0 \times ab}{b}}{\frac{b-a}{b}}$$ [Dividing both by b]

$$= \frac{4\pi\epsilon_0 \times a}{1 - \frac{a}{b}}$$

$$= \frac{4\pi\epsilon_0 \times a}{1 - \frac{a}{\infty}}$$ [$b = \infty$]

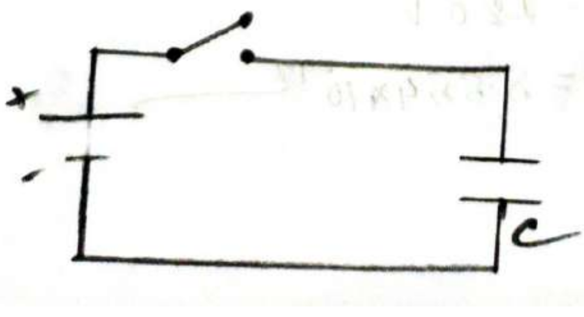
$$= \frac{4\pi\epsilon_0 \times a}{1 - \frac{a}{\infty}}$$

$$= \frac{4\pi\epsilon_0 \times a}{1 - 0}$$

$$\therefore \boxed{C = 4\pi\epsilon_0 \times a}$$ → This is the Capacitance for isolated sphere.

Problem: 2

⇒ The Capacitor in the adjacent Fig. has a capacitance of 25μF and is initially uncharged. The battery provides a potential difference of 120 V. After switch S is closed, how much charge will pass through it?



Topic Name : Day : Time : Date : / /

⇒ is given:—

$$C = 25\mu F = 25 \times 10^{-6} F$$

$$V = 120 V$$

We know:—

$$C = \frac{q}{V}$$

$$q = CV$$

$$= 25 \times 10^{-6} \times 120$$

$$= 3 \times 10^{-3} C$$

Ans

Problem: 3

⇒ A parallel-plate capacitor has circular plates of 8.20 cm radius and 1.30 mm separation.

(a) Calculate the capacitance.

(b) Find the charge for a potential difference of 120 V.

⇒ is given:—

$$r = 8.20 cm = 8.20 \times 10^{-2} m$$

$$d = 1.30 mm = 1.30 \times 10^{-3} m$$

$$V = 120 V$$

$$\epsilon_0 = 8.854 \times 10^{-12}$$

Topic Name : Day : Time : Date : / /

(a)

We know:—

$$C = \frac{\epsilon_0 (\pi r^2)}{d}$$

$$= \frac{8.854 \times 10^{-12} \times (\pi \times (8.2 \times 10^{-2})^2)}{1.30 \times 10^{-3}}$$

$$= 1.43 \times 10^{-10} F$$

Ans:—

(b)

We know:

$$C = \frac{q}{V}$$

$$q = CV$$

$$= 1.43 \times 10^{-10} \times 120$$

$$= 1.72 \times 10^{-8} C$$

Problem:- 04:-

⇒ The plates of a spherical capacitor have radii 38mm and 40mm.

(a) Calculate the capacitance.

(b) What must be the plate area of a parallel-plate capacitor with the same plate separation and capacitance.

⇒ is given:-

$$a = 38\,mm = 38 \times 10^{-3}\,m$$
$$b = 40\,mm = 40 \times 10^{-3}\,m$$

(a)

We know:-

$$C = \frac{4\pi\epsilon_0 ab}{b-a}$$

$$= \frac{4 \times 8.854 \times 10^{-12} \times \pi \times 38 \times 10^{-3} \times 40 \times 10^{-3}}{(40-38) \times 10^{-3}}$$

$$= 8.45 \times 10^{-11}\ F$$

Topic Name : Day : Time : Date : / /

(b)

$d = b - a = 2\times10^{-3} m$

we know:-

for a parallel plate capacitor:-

$$C = \frac{\epsilon_0 A}{d}$$

$$A = \frac{Cd}{\epsilon_0}$$

$$= \frac{8.45\times10^{-11} \times 2\times10^{-3}}{8.854\times10^{-12}}$$

$$= 10.087\times10^{-3} m^2$$

Ans:-

Problem: 6:-

=> You have two flat metal plates, each of area $1m^2$ with which to construct a parallel plate capacitor.

(a) If the capacitance of the device is to be 1F What must be the separation between the plates.

(b) Could this capacitor actually be constructed?

P.T.O

Topic Name: Day: Time: Date: / /

⇒ is given:-

$A = 1m^2$

$C = 1F$

$\epsilon_0 = 8.854 \times 10^{-12}$

(a)

we know:-

$$C = \frac{\epsilon_0 A}{d}$$

$$d = \frac{\epsilon_0 A}{C}$$

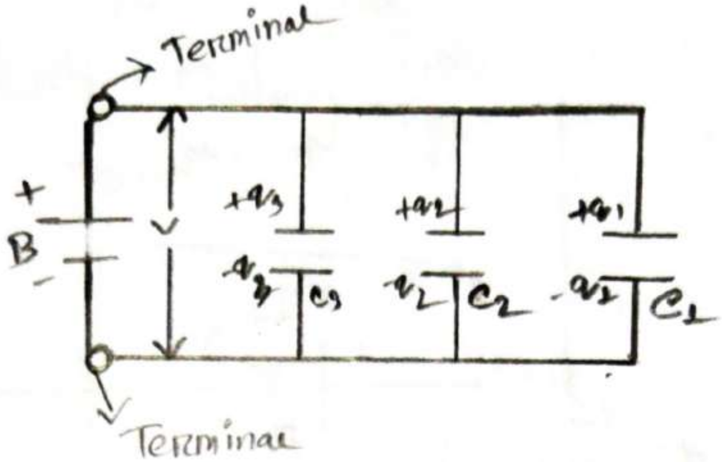
$$= \frac{8.854 \times 10^{-12} \times 1}{1}$$

$$= 8.854 \times 10^{-12} m$$

(b)

It is not possible to construct a capacitor by the separation distance. $d = 8.854 \times 10^{-12} m$ because d value is less than the minimum size of atom.

Capacitors in parallel Combination:-



Charge on each capacitor:-

$q_1 = c_1 V$

$q_2 = c_2 V$

$q_3 = c_3 V$

So, the total charge on parallel combination is:-

$$q = q_1 + q_2 + q_3$$

$$q = (c_1 + c_2 + c_3) \times V$$

Now:- The equivalent capacitance is:-

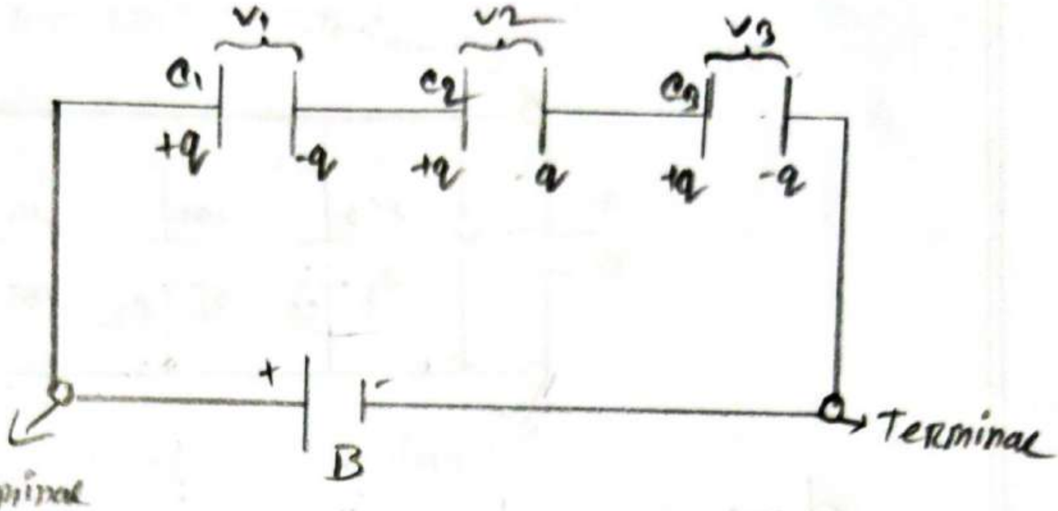
$$C_{eq} = \frac{q}{V}$$

$$C_{eq} = \frac{(c_1 + c_2 + c_3) V}{V}$$

$$\boxed{C_{eq} = c_1 + c_2 + c_3}$$

Topic Name : Day : Time : Date : / /

Capacitors in Series Combination:—



Potential diffrence of each actual capacitor:—

$V_1 = \frac{q}{C_1}$; $V_2 = \frac{q}{C_2}$; $V_3 = \frac{q}{C_3}$

So, the total potential diffrence $= V$ is;

$$V = V_1 + V_2 + V_3$$
$$= q\left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_2}\right)$$

Now: The equivalent capacitance is:—

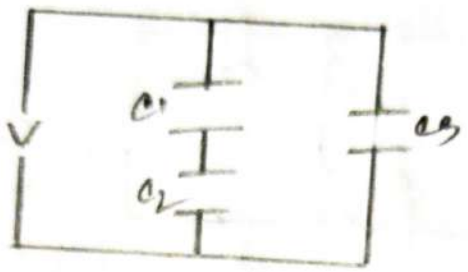
$$C_{eq} = \frac{q}{V}$$
$$= \frac{q}{q\left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}\right)}$$

$$\boxed{\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}}$$

Topic Name : Day : Time : Date : / /

Problem 10:—

⇒ In the adjecent Figure, Find the equivalent capacitance of the combination. Assume that $C_1 = 10\mu F$, $C_2 = 5\mu F$, $C_3 = 4\mu F$



⇒ is given—

$C_1 = 10\mu F$

$C_2 = 5\mu F$

$C_3 = 4\mu F$

here, C_1 And C_2 are in Series:—

$$\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2}$$

$$= \frac{1}{10} + \frac{1}{5}$$

$$= \frac{3}{10}$$

$$C_s = \frac{10}{3}\ \mu F$$

And C_s And C_3 Are in parallel:—

$$C_{eq} = C_s + C_3$$

$$= \frac{10}{3} + 4$$

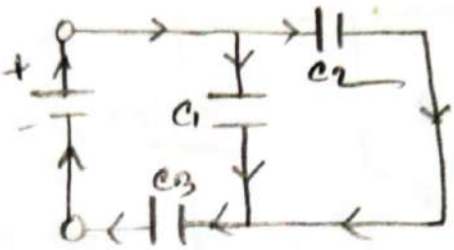
$$= \frac{22}{3}\ \mu F$$

$$= 7.33 \times 10^{-6} F$$

Topic Name : Day : Time : Date : / /

Problem 11:—

⇒ In the adjacent figure, find the equivalent capacitance of the combination. Assume that $C_1 = 10\mu F$, $C_2 = 5\mu F$, $C_3 = 4\mu F$



⇒ is given:—

$C_1 = 10\mu F$
$C_2 = 5\mu F$
$C_3 = 4\mu F$

here, C_1 and C_3 are in Series:—

$$\frac{1}{C_{eq_s}} = \frac{1}{C_1} + \frac{1}{C_3}$$

$$= \frac{1}{10} + \frac{1}{4}$$

$$= \frac{7}{20}$$

$$\therefore C_{eq_s} = \frac{20}{7}\mu F$$

Now, C_{eq_s} and C_2 are in Parallel:—

$$\therefore C_{eq} = C_{eq_s} + C_2$$

$$= \frac{20}{7} + 5$$

$$= \frac{55}{7}\mu F$$

$$= 7.85 \times 10^{-6} F$$

Topic Name : Day : Time : Date : / /

⇒ is given:—

$C_1 = 10\mu F$

$C_2 = 5\mu F$

$C_3 = 4\mu F$

here, C_1 and C_2 are in parallel :—

$$C_{eq_p} = C_1 + C_2$$
$$= 10 + 5$$
$$= 15\mu F$$

Now:— C_{eq_p} and C_3 are in Series:—

$$\frac{1}{C_{eq_s}} = \frac{1}{C_{eq_p}} + \frac{1}{C_3}$$
$$= \frac{1}{15} + \frac{1}{4}$$
$$= \frac{19}{60}$$

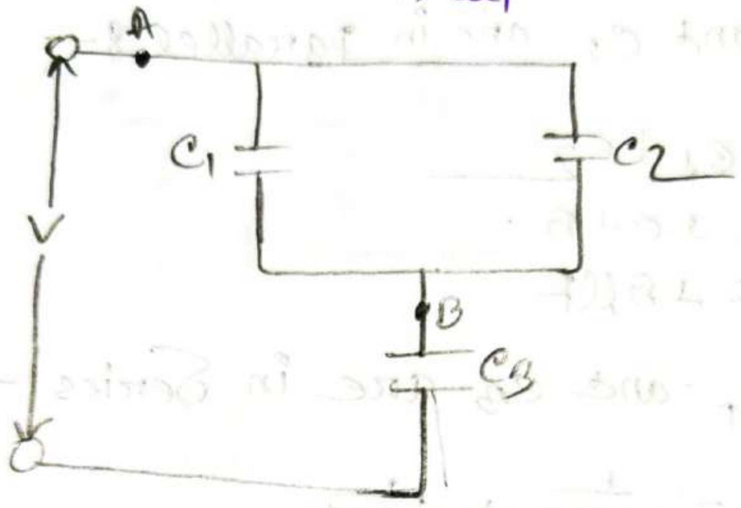
$$\therefore C_{eq_s} = \frac{60}{19} \mu F$$
$$= 3.157 \times 10^{-6} F$$

Ans:—

Topic Name : Day : Time : Date : / /

Problems 25.02

Find the equivalent capacitance for the combination of capacitances shown in figure, Assume $C_1 = 12\mu F$, $C_2 = 5.30\mu F$, $C_3 = 4.5\mu F$



here,

$C_1 = 12\mu F$

$C_2 = 5.30\mu F$

$C_3 = 4.5\mu F$

here, C_1 and C_2 are in parallel:—

$$C_{12} = 12 + 5.30$$
$$= 17.3\mu F$$

Now:— C_{12} and C_3 are in Series:—

$$\frac{1}{C_{123}} = \frac{1}{C_{12}} + \frac{1}{C_3}$$
$$= \frac{1}{17.3} + \frac{1}{4.5}$$
$$= \frac{436}{1557}$$

$C_{123} = 3.57\mu F$ Ans:—

Topic Name : Day : Time : Date : / /

Energy Stored in an Electric Field:—

⇒ Suppose that,

at a given instant, a charge q' hasbeen transferred from one plate of a capacitor to other. The potential difference V' between the plates at that instant will be $\frac{q'}{C}$.

If the extra increment of charge dq' is then transferred, the increment of work required will be

$$dW = V'dq'$$

$$\boxed{dW = \frac{q'}{C} \times dq'}$$

The work required to bring the total capacitor charge up to a final value q is

$$W = \int_{q'=0}^{q'=q} dW = \frac{1}{C}\int_0^q q'dq' = \frac{q^2}{2C}$$

The work stored as potential energy U in the capacitor,

$$U = W = \frac{q^2}{2C} = \frac{(CV)^2}{2C} = \frac{C^2V^2}{2C} = \frac{1}{2}CV^2$$

$$\therefore \boxed{U = \frac{q^2}{2C} = \frac{1}{2}CV^2}$$